MACROECONOMICS

MACRO-23

Time: 5 minutes

Outcome: Apply the quantity equation, velocity, money growth, price levels, and inflation.

Difficulty: ●●○ Intermediate

Quantity Theory of Money

THE CORE IDEA

The quantity equation MV = PY links the money stock and velocity to nominal output. Its growth-rate form separates growth in money and velocity from growth in real output and the price level. When velocity is stable and real output is determined by productive capacity, persistent money growth above real-output growth produces long-run inflation rather than permanently higher real output.

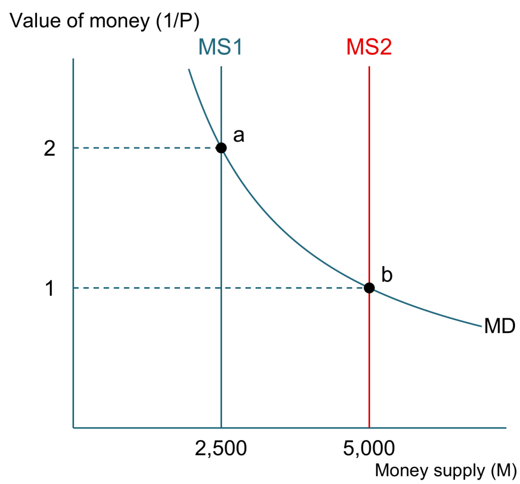
HOW TO RECOGNIZE IT

- The problem uses MV = PY or relates money growth to inflation.
- Velocity measures how often money supports transactions during a period.
- With velocity and real output fixed, more money raises the price level.
- The question distinguishes a nominal price change from a lasting real-output change.

WATCH OUT

Do not assume money growth permanently raises real output. The quantity theory connects persistent money growth to inflation, especially in the long run.

WORKED EXAMPLE: MONEY GROWTH AND INFLATION



In this clean quantity-theory case, velocity V and real output Y are fixed. Money supply doubles from 2,500 (MS1) to 5,000 (MS2). The value of money falls from 2 at a to 1 at b. Since P is its reciprocal, P rises from 1/2 = 0.50 to 1/1 = 1.00: a 100% increase. The graph shows value of money (1/P), not velocity V in MV = PY.

CHECK YOURSELF

In MV = PY, what does P represent?

READY? **Return to the game and master this concept.**